SACHIIN M

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Education

B.E. Computer Science & Engineering (AIML) 2023–2027
KPR Institute of Engineering and Technology, Coimbatore, India CGPA: 7.5 / 10
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Computer Networks

Work Experience

Project Intern — Payoda, Coimbatore Feb – Jul 2026
FastAPI · Node.js · React · Playwright · faster-whisper · LLMs

- Built a complete **e-commerce backend** using **FastAPI**, implementing structured routing, Pydantic schemas, and **scalable**, fault-tolerant service layers with a **React** admin dashboard for order and inventory management.
- Deployed a **WebSocket-based streaming architecture** for real-time speech transcription using **faster-whisper**, achieving zero API cost.
- Engineered **Node.js automation scripts** via **Playwright** for end-to-end testing across internal modules (AI TestGen, SnapTest, ScriptWeaver), debugging and resolving integration issues across distributed test environments; participated in **code reviews** to uphold code quality and maintainability standards.
- Architected a **multi-agent AI system** optimizing LLM token utilization and dynamic context lengths for distributed inter-agent workflows, contributing directly to internal **GenAI** tooling for development productivity.

Software Developer Intern — Zettaone Technologies, Bengaluru Jul – Aug 2025
Flask · React Native · OpenCV · EasyOCR · YOLOv8

- Architected a **PCB component deviation detection** system using **YOLOv8 + EasyOCR**, identifying manufacturing defects in real time via a Flask-based CV pipeline in a production environment, with results persisted for durable storage and downstream audit.
- Engineered **VISYRE** (React Native + Flask) for QR/barcode decoding and OCR-based text extraction, eliminating manual data entry; participated in **agile sprints** with daily stand-ups and retrospectives, using **Git** feature branches to manage parallel workstreams.
- Built **Climora**, a cross-platform React Native app for real-time weather and soil insights via geolocation APIs, backed by a **microservices-style**, scalable Flask backend with an emphasis on **operational excellence** and reliability.

Research Intern — Providence University, Taiwan Oct – Nov 2025
CNN · PyTorch · Grad-CAM · Medical Imaging

- Trained a **CNN-based pneumonia classifier** on **7,750 chest X-ray images**, achieving **92.44% overall accuracy**, **0.94 F1-score** on pneumonia class, and **0.93 weighted precision**; leveraged GPU-accelerated training infrastructure, reducing wall-clock time by 3× vs. local hardware.
- Integrated **Grad-CAM** heatmaps to localise prediction regions and generated **automated PDF medical reports** per patient for interpretable clinical diagnostic output.
- Applied SMOTE, rotation, flipping, zoom, and contrast augmentation to resolve 4:1 class imbalance; followed a **Scrum workflow** — weekly sprint planning, tasks tracked via GitHub Issues — improving team velocity by 20%.

Projects

Morpheus — Natural Language to Blender via MCP (GenAI Tooling) 2025
React · Ollama · MCP · Blender Python API (bpy) · FastAPI

- Integrated **Model Context Protocol (MCP)** with Blender to enable LLM-driven scene control allowing **Ollama** (local LLM) to directly manipulate 3D objects, materials, and lighting through natural language prompts, showcasing practical **GenAI** application development.
- Built a **React** web dashboard for prompt input, live scene preview, and tool-call history, communicating with the **FastAPI** backend over REST endpoints.

VoiceScript — Real-time Speech Transcription System 2025
React · FastAPI · WebSocket · faster-whisper · Silero VAD

- Built a **self-hosted real-time transcription system** using **faster-whisper** (large-v3-turbo) for accurate speech-to-text, achieving **sub-second streaming latency** with zero API cost.
- Developed a **React** frontend with a live-updating transcript view, consuming streamed **WebSocket** data to render partial and finalized captions in real time.
- Integrated **Silero VAD** for voice activity detection and **LocalAgreement algorithm** for stable partial transcripts, eliminating word-level jitter across live audio streams.

Skills

Languages	Python, JavaScript, Java, SQL, Node.js
Frameworks	React, React Native, FastAPI, Flask, PyTorch, TensorFlow, Angular, Playwright
System Design	Microservices, REST APIs, Message Queues, Fault Tolerance, Load Balancing, Caching, Scalable Systems
ML / GenAI	OpenCV, YOLOv8, EasyOCR, Grad-CAM, faster-whisper, Silero VAD, Multi-Agent Systems, LLMs, MCP
Tools	Git, Docker, Streamlit, VS Code, Blender (bpy), Ollama, CrewAI
Databases	Postgres, MySQL, MongoDB
Practices	Agile/Scrum, Sprint Planning, Code Review, GitHub Issues, Operational Excellence